

No. of Functional Features Included in the Solution

Date	01 July 2026
Team ID	
Project Name	Rising Waters: Smart AI Flood Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S. No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form Interface	Web form capturing user parameters such as annual, seasonal rainfall patterns and cloud visibility.	Web Dashboard Frontend (templates/)	Done	1.0 Marks
2	Dynamic Feature Standardization	Pipeline utilizing Scikit-Learn to scale incoming live meteorological inputs to match training distributions.	Data Processing Module (preprocess.py)	Done	0.5 Marks
3	Multi-Model Evaluation Engine	Comparative infrastructure benchmarking Decision Trees, Random Forests, KNN, and XGBoost models.	Machine Learning Pipeline (train.py)	Done	1.0 Marks
4	High-Accuracy Model Inference	Live pipeline loading the serialized XGBoost engine (floods.save) to classify data with 96.55% accuracy.	Model Execution Kernel (app.py)	Done	1.0 Marks

5	Dynamic Result Presentation Layer	Dedicated response routes mapping classifications into immediate visual indicators (Flood / No Flood Risk panels).	UI Template Component (views/)	Done	0.5 Marks
6	Distributed Deployment Runtime	Production-ready manifest files optimized for hosting the prediction web stack natively on IBM Cloud.	Cloud Platform Deployment Container	Done	0.5 Marks
7	Real-Time Telemetry API Integration	Webhook scheduler designed to query and auto-fill live local climate feeds without needing manual form entry.	Automation Pipeline Engine	In Progress	0.25 Marks
8	Interactive Geolocation Map Layers	Map dashboard layer displaying historically documented flood-prone regions and critical warning areas.	Spatial Visualization Scripting	Pending	0.25 Marks

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	6
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Interactive Input Web Form, Flood / No Flood Chance visual result pages.
2	Backend / Logic	3	Dynamic rainfall feature scaling, Multi-algorithm verification pipeline, High-accuracy XGBoost live predictive engine.

3	Database / Storage	1	Serialization mapping and local model asset storage container (floods.save).
4	API / Integration	1	Automated third-party weather data fetching webhook synchronization layer (In Progress).
5	Security / Authentication	1	Input validation sanitization scripts to prevent malformed telemetry injection payloads.